

Software Requirements Specification

Academic Performance Early-Warning System

Group Members

Shaiza Hamid – 24F-0585

Ayesha Naeem – 24F-0586

Department of Computer Sciences

FAST – National University of Computer & Emerging Sciences

Chiniot-Faisalabad Campus

2026

Anti-Plagiarism Declaration

This is to declare that the above publication produced under the:

Title: Academic Performance Early-Warning System

is the sole contribution of the author(s) and no part hereof has been reproduced on an as-it-is basis (cut and paste) which can be considered as Plagiarism. All referenced parts have been used to argue the idea and have been cited properly. We will be responsible and liable for any consequence if violation of this declaration is determined.

Date: 13th March, 2026

Student 1

Name: Shaiza Hamid

Signature: _____

Student 2

Name: Ayesha Naeem

Signature: _____

Version: 1.0

Date: 13th March 2026

Prepared by: Shaiza Hamid, Ayesha Naeem

Intellectual Property Right Declaration

This is to declare that the work under the:

Title: Academic Performance Early-Warning System carried out in partial fulfillment of the requirements of: BS Course Project

Is the sole property of the National University of Computer and Emerging Sciences and is protected under intellectual property right laws. It can only be considered/used for purposes like extension for further enhancement, product development, or adoption for commercial/organizational usage, with the permission of the University.

This above statement applies to all Users and faculty.

Date: 13th March 2026

Student 1:

Name: Shaiza Hamid

Signature:

Student 2:

Name: Ayesha Naeem

Signature:

Dedication

This work is dedicated to our beloved parents, whose endless support, prayers, and encouragement have been our source of strength throughout this journey.

We also dedicate this project to our teachers, who guided us with patience, wisdom, and sincerity.

Thank you for believing in us.

Abstract

This project presents the design and analysis of an Academic Performance Early-Warning System, a software system developed to address the challenges faced in academic performance monitoring within higher education institutions. The primary objective of the system is to provide an efficient and user-friendly platform that allows academic advisors, faculty, and administrators to monitor student performance indicators while improving the overall management and accessibility of academic data.

The proposed system aims to automate existing manual processes and enhance decision-making through structured data handling and system integration. The system supports different types of users including Students, Academic Advisors, Faculty Members, and System Administrators, each with specific functionalities and access privileges. Core features of the system include risk classification, automated alert generation, real-time dashboards, intervention logging, and reporting and analytics.

The system is designed following standard Software Engineering practices, including requirements analysis, system modelling, and UML-based design. Various diagrams such as Use Case Diagrams, Domain Models, System Sequence Diagrams, and Sequence Diagrams are used to represent system behaviour and interactions clearly. Overall, the proposed solution aims to improve efficiency, transparency, and usability while ensuring scalability, security, and maintainability of the system.

Table of Contents

1 Introduction	1
1.1 Purpose	1
1.2 Intended Audience	1
1.3 Project Scope	1
1.4 References	2
2 Vision Document	3
2.1 Problem Statement	3
2.2 Business Opportunities	3
2.3 Objectives	4
2.4 Scope	5
2.5 Constraints	6
2.6 Stakeholder and User Description	6
2.6.1 Market Demographics	6
2.6.2 User Environment	6
2.6.3 Stakeholder Profiles	7
2.6.4 Stakeholder Summary	10
3 System Requirements Specification	11
3.1 System Features	11
3.2 Functional Requirements	12
3.2.1 Product / Organizational Requirements	14
3.2.2 Business Requirements	14
3.3 Non-Functional Requirements	15
4 Requirements Analysis Phase Diagrams	16
4.1 Use Case Diagram	16
4.2 Expanded Use Cases	17
4.2.1 User Registration Use Cases	17
4.2.2 User Login Use Cases	19
4.2.3 Student Dashboard Use Case	21
4.2.4 Risk Classification Use Cases	23
4.2.5 Advisor Alert Use Cases	25
4.2.6 Intervention Logging Use Case	27
4.2.7 Report Generation Use Case	29
4.2.8 System Configuration Use Case	31
4.3 System Sequence Diagrams (SSD)	33
4.3.1 Student Registration SSD	33
4.3.2 User Login SSD	34
4.3.3 Risk Assessment SSD	35
4.3.4 Alert Notification SSD	36
4.3.5 Advisor Dashboard SSD	37
4.3.6 Intervention Logging SSD	38

4.3.7 Report Generation SSD	39
4.3.8 System Configuration SSD	40
5 Design Phase Artifacts	41
5.1 Domain Model	41
5.2 Sequence Diagram	43

Deliverable-03

6 Detailed Design & Modeling Artifacts	46
6.1 System Sequence Diagrams (SSD)	47
6.2 Operation Contracts	48
7 Structural Design	57
7.1 Class Diagram	57
7.2 Package Diagram	58
8 Interaction Design	59
8.1 Communication Diagram	59
9 Behavioral Modeling	60
9.1 State Machine Diagram	60
9.2 Activity Diagram	61
10 Data Flow Modeling	62
10.1 Data Flow Diagram (DFD)	62

List of Tables

Table 1: Problem Statement	3
Table 2: Stakeholder Summary	6
Table 3: Academic Advisor	8
Table 4: Faculty Member	9
Table 5: Student	9
Table 6: Department Administrator	10
Table 7: Functional Requirements	12
Table 8: Business Requirements	14
Table 9: Non-Functional Requirements	15
Table 10: User Registration	17
Table 11: User Login	19
Table 12: View Student Dashboard	21
Table 13: Risk Classification	23
Table 14: Send Alert Notification	25
Table 15: Log Intervention	27
Table 16: Generate Report	29
Table 17: Configure System Thresholds	31

List of Figures

Figure 1: Use Case Diagram – Academic Performance Early-Warning System	16
Figure 2: Student Registration SSD	33
Figure 3: User Login SSD	34
Figure 4: Risk Assessment SSD	35
Figure 5: Alert Notification SSD	36
Figure 6: Advisor Dashboard SSD	37
Figure 7: Intervention Logging SSD	38
Figure 8: Report Generation SSD	39
Figure 9: System Configuration SSD	40
Figure 10: Domain Model	41
Figure 11: Complete System Sequence Diagram	43

1 Introduction

In many academic institutions, student performance problems are usually noticed only after serious decline has already happened — including repeated failures, low attendance, or students dropping courses. Most existing academic systems only store grades and attendance records without analyzing this data to identify early warning signs or support timely academic help.

At present, advisors and teachers mainly depend on manual observation or delayed reports, making it difficult to identify students whose performance is slowly decreasing due to heavy workload, poor time management, or academic pressure. This late identification leads to delayed support, poor academic results, and a higher risk of students leaving their studies.

The proposed Academic Performance Early-Warning System aims to solve these problems by providing a software system that regularly monitors academic indicators such as attendance, test results, and student workload. The system uses predefined rules to analyze this data and classify students into different risk levels, allowing institutions to shift from a reactive to a preventive model of academic performance management.

1.1 Purpose

This document describes the complete software requirements for the Academic Performance Early-Warning System. It is intended to establish a shared understanding of the system goals, functional behavior, constraints, and quality attributes among all project stakeholders including developers, supervisors, and end users.

1.2 Intended Audience

This document is intended for the development team, project supervisor, academic advisors, faculty members, and institutional administrators who will use or oversee the system. It also serves as a reference for testers and maintenance personnel.

1.3 Project Scope

The system will be developed as a web-based application for higher education institutions. It will continuously monitor student academic performance indicators, classify students into risk levels, and send alerts to responsible stakeholders. The system integrates with existing Student Information Systems (SIS) and Learning

Management Systems (LMS) but does not replace them. Features such as automated grade modification and enrollment changes are explicitly out of scope.

1.4 References

The following references were consulted during the preparation of this document:

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- FAST-NU Academic Information System Documentation
- Sample SRS Document provided by course instructor
- Deliverable-01 Outlines (Deliverable-02_outlines.pdf)

2 Vision Document

2.1 Problem Statement

The following table summarizes the core problem, its effects, impact, and the proposed solution for the Academic Performance Early-Warning System.

Table 1: Problem Statement

Category	Description
Problem	Student performance deterioration is detected only after reaching a critical stage. Existing systems only store grades/attendance and lack analytical capabilities. Advisors rely on manual, time-consuming reviews prone to oversight. High advisor-to-student ratios make individual monitoring impractical.
Affects	Undergraduate students, academic advisors, faculty, and institutional administration. Erodes academic support quality and reduces institutional retention rates.
Impact	Delayed support leads to poor academic results and higher dropout risk. Missed early warnings result in irreversible academic decline for at-risk students.
Solution	A software system that continuously monitors academic indicators (attendance, grades, workload) and classifies students into risk levels using predefined rules, enabling timely advisor intervention and shifting institutions from reactive to preventive academic management.

2.2 Business Opportunities

There is a significant opportunity to improve student retention and academic outcomes through an intelligent monitoring system. Educational institutions increasingly recognize that retaining students and improving graduation rates has direct financial and reputational benefits. By investing in this system, universities can reduce dropout rates, improve advisor efficiency, and demonstrate commitment to student success. The system also provides data-driven insights that can inform curriculum design and resource allocation decisions, creating long-term institutional value.

2.3 Objectives

Following are the objectives of the Academic Performance Early-Warning System:

- Develop a software system capable of continuously monitoring key academic performance indicators including attendance rates, assignment completion, quiz and test scores, and overall workload.
- Implement a risk classification engine that categorizes students into risk levels (Low, Medium, High) based on predefined threshold rules.
- Provide advisors and faculty with a real-time dashboard displaying student risk statuses and trends, enabling timely and targeted intervention.
- Generate automated alerts and notifications when a student's risk level escalates.
- Maintain a secure, role-based access system that protects student data privacy while enabling appropriate stakeholders to access relevant information.
- Shift the institution's academic performance management from a reactive approach to a proactive, preventive model.

2.4 Scope

The system will be developed as a web-based application integrated with the institution's existing academic information systems. It encompasses modules for data collection, risk analysis, dashboard visualization, and notification management. The system will cover undergraduate students and will interface with existing course management and student information systems. It is not intended to replace existing academic management tools but will function as a supplementary analytical layer. Features such as automated grade modification and enrollment changes are out of scope.

2.5 Constraints

- Technical: Must integrate with existing legacy academic systems that may have limited API capabilities.
- Data Privacy: Must comply with applicable data protection regulations; access must be strictly role-based and audited.
- Resource: Project must be completed within the academic semester with the available development team.
- Regulatory: All data handling must align with educational data privacy laws applicable to the institution.

- Operational: Must be available during regular academic hours with minimal downtime.

2.6 Stakeholder and User Description

This section describes the key stakeholders and users who have an interest in the system. Understanding their needs and roles is essential for defining appropriate features and access controls.

2.6.1 Market Demographics

The primary market consists of higher education institutions with medium-to-large student populations where advisor-to-student ratios exceed 1:50, making manual monitoring impractical. Secondary markets include vocational training institutions and community colleges seeking to improve retention rates.

2.6.2 User Environment

Academic advisors and faculty will access the system via web browsers on institutional desktops or laptops. The system must be compatible with Chrome, Firefox, and Edge. Students will access it via both desktops and mobile devices, requiring a fully responsive design. No specialized software beyond a modern web browser is required.

2.6.3 Stakeholder Profiles

Table 3: Academic Advisor

Representatives	Academic Advisors
Description	Primary users who monitor student risk statuses and coordinate support.
Type	Internal Stakeholder
Responsibility	Monitor assigned students' risk dashboards daily, respond to escalation alerts, schedule intervention meetings, and log outcomes.
Success Criteria	Timely identification and intervention for all high-risk students within 48 hours of risk escalation.
Involvement	Daily system user; provides feedback for threshold configuration.
Comments / Issues	Requires intuitive interface due to varying technical proficiency levels.

Table 4: Faculty Member

Representatives	Course Instructors / Faculty
Description	Provide academic data inputs and may view risk indicators for their enrolled students.
Type	Internal Stakeholder
Responsibility	Confirm attendance records, upload grade data, and review student risk indicators for courses they teach.
Success Criteria	Accurate and timely data submission enabling reliable risk analysis.
Involvement	Periodic system access for data validation and course-level review.
Comments / Issues	Access restricted to students enrolled in their own courses.

Table 5: Student

Representatives	Enrolled Undergraduate Students
Description	Subjects of the monitoring system who may view their own performance data.
Type	External Stakeholder
Responsibility	Review personal risk status, act on recommendations, and seek advisor support when notified.
Success Criteria	Improved awareness of academic standing and timely engagement with support resources.
Involvement	Self-service portal access; no administrative capabilities.
Comments / Issues	Cannot view other students' data; privacy is paramount.

Table 6: Department Administrator

Representatives	Department Heads / Senior Academic Staff
Description	Oversee program-wide performance trends and configure system parameters.
Type	Internal Stakeholder
Responsibility	Configure risk thresholds, review aggregate reports, and make strategic academic decisions based on system insights.
Success Criteria	Measurable improvement in departmental retention and graduation rates.

Involvement	Periodic access for reporting and configuration management.
Comments / Issues	Does not have access to individual student intervention records.

2.6.4 Stakeholder Summary

Table 2: Stakeholder Summary

Name	Description	Responsibility
Development Team	Involved in designing and developing the Early-Warning System.	Fully involved in system development, requirement engineering, research, and documentation.
Supervisor Team	Supervise and guide the development team throughout the project lifecycle.	Provide technical direction, review progress, ensure timely delivery, and validate product features.
Academic Advisors	Primary users who monitor student risk dashboards and initiate interventions.	Monitor assigned students, respond to alerts, log intervention actions.
Faculty Members	Instructors who provide course-level data inputs to the system.	Upload or confirm attendance and grade data; view risk status of enrolled students.
Students	Subjects of monitoring who can view their own academic performance indicators.	View personal risk status and act on improvement recommendations.
Administrators	Configure system thresholds and view aggregate department-wide reports.	Set risk parameters, review program-level performance, generate reports.

3 System Requirements Specification

This section defines the system features, functional requirements, and non-functional requirements. These requirements form the basis for system design, development, and acceptance testing.

3.1 System Features

- **Academic Data Monitoring:** Continuously collects and updates academic performance data including attendance, assignment grades, quiz scores, and exam results.
- **Risk Classification Engine:** Analyzes collected data using predefined rule-based algorithms to assign each student a risk level: Low, Medium, or High. Thresholds are configurable by administrators.
- **Real-Time Dashboard:** Provides a role-appropriate dashboard displaying student risk statuses, performance trends, and summary statistics tailored to each user type.
- **Automated Alert System:** Generates email and in-system notifications to advisors and students when risk levels change or escalate.
- **Intervention Logging:** Allows advisors to record details of any intervention, including date, type, and outcome, accessible only to authorized personnel.
- **Reporting and Analytics:** Generates scheduled and on-demand reports on performance trends, risk distribution, and intervention effectiveness.

3.2 Functional Requirements

The functional requirements define the specific behaviors the system must perform. Each requirement describes the action the system must take and the expected outcome.

Table 7: Functional Requirements

ID	Requirement
FR-01	The system shall retrieve student academic data from the institutional database at intervals not exceeding 24 hours during the academic term.
FR-02	The system shall calculate a risk score based on weighted indicators: attendance (30%), assignment completion (20%), quiz performance (20%), and examination performance (30%).

ID	Requirement
FR-03	The system shall classify each student into a risk category (Low, Medium, or High) based on their computed risk score and configurable threshold values.
FR-04	The system shall display a student risk dashboard to authorized advisors, showing risk level, trend (improving/stable/declining), and key contributing factors.
FR-05	The system shall send automated notifications to the responsible advisor and the student within 24 hours of a risk level escalation.
FR-06	The system shall allow advisors to search and filter students by name, course, risk level, or enrollment year.
FR-07	The system shall allow students to log in and view their own performance indicators, risk status, and personalized improvement recommendations.
FR-08	The system shall allow administrators to configure risk threshold values and indicator weightings without requiring system redeployment.
FR-09	The system shall maintain an audit log of all data access events, recording user, timestamp, and action performed.
FR-10	The system shall generate a monthly performance report for department administrators summarizing risk distribution, intervention counts, and performance trends.

3.2.1 Product / Organizational Requirements

The system shall integrate seamlessly with the institution's existing Student Information System (SIS) and Learning Management System (LMS), supporting RESTful APIs and CSV data imports. It must comply with institutional data governance policies and receive IT security committee approval prior to deployment. All development shall adhere to the organization's software development lifecycle standards.

3.2.2 Business Requirements

Table 8: Business Requirements

ID	Requirement
BR-01	Reduce the average time to identify an at-risk student from several weeks (manual) to within 48 hours of data availability.

ID	Requirement
BR-02	Improve academic advisor efficiency by reducing manual data review time by at least 40%.
BR-03	Contribute to an improvement in the institution's student retention rate through timely intervention.
BR-04	Support the institution's strategic goal of increasing graduation rates via data-driven academic insights.
BR-05	Scale to support up to 120% of current student enrollment without performance degradation.

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes the system must satisfy — governing how the system performs its functions rather than what functions it performs.

Table 9: Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Dashboard requests shall respond within 3 seconds for up to 200 concurrent users. Report generation shall complete within 10 seconds.
NFR-02	Reliability & Availability	The system shall maintain a minimum uptime of 99.5% during the academic term. Maintenance shall be communicated 24 hours in advance.
NFR-03	Security	All data encrypted in transit using TLS 1.2+. Authentication via institutional SSO. Role-based access control enforced. Audit trail retained for 3 years.
NFR-04	Scalability	Architecture shall support horizontal scaling for up to 20,000 students. Database optimized for read-heavy dashboard workloads.
NFR-05	Usability	Interface shall comply with WCAG 2.1 Level AA. New advisors operational after a one-hour orientation. Responsive from 375px to 1920px.
NFR-06	Maintainability	Codebase with comprehensive documentation. Automated test suites covering at least 80% of core logic. Loosely coupled modules.
NFR-07	Portability	Deployable on standard cloud infrastructure (AWS, Azure, or on-premises Linux). Compatible with major browsers without plugins.

ID	Category	Requirement
NFR-08	Data Integrity	Incoming data validated; non-conforming records flagged. Historical data stored in immutable format for longitudinal analysis.

Trigger	A new user (Student / Advisor / Faculty / Administrator) initiates registration on the platform.
Pre-Condition	User has a valid institutional email address. User is not already registered in the system. System is operational and accessible.
Post-Condition	User account is created with an appropriate role assigned. User credentials are stored securely in the database. User receives a confirmation notification.
Normal Flow	1. User navigates to the registration page. 2. User selects their role (Student / Advisor / Faculty / Administrator). 3. User enters personal information (name, email, student/staff ID, password). 4. System validates the entered information. 5. System assigns appropriate user role based on selection. 6. System creates user account and generates a unique user ID. 7. System sends a verification email. 8. User verifies account through the link. 9. System redirects user to the appropriate dashboard.
Alternate Flow	AF-1 (Email Already Exists): System detects duplicate email → displays error message → prompts user to log in or use password recovery. AF-2 (Incomplete Information): User submits form with missing fields → system highlights required fields → user completes information. AF-3 (Verification Timeout): User does not verify within 24 hours → system sends reminder → account remains inactive until verified.
Special Requirements	Password must meet institutional security standards. Role assignment must be verified by system administrator for privileged roles.
Frequency of Use	High (Multiple times during enrollment periods)

4.2.2 User Login Use Cases

Table 11: User Login

Use Case ID	UC-002
Use Case Name	User Login
Priority	Critical
Trigger	Registered user attempts to access the system.

Pre-Condition	User has a registered account. User account is active and verified. System is operational.
Post-Condition	User is authenticated and logged into the system. User session is created with appropriate permissions. User is redirected to role-specific dashboard. Login attempt is logged for security audit.
Normal Flow	1. User navigates to the login page. 2. User enters email/username and password. 3. System validates credentials against the database. 4. System checks account status (active/suspended). 5. System creates an authenticated session. 6. System logs login activity (timestamp, IP address, device). 7. System redirects user to the appropriate portal based on role.
Alternate Flow	AF-1 (Invalid Credentials): System detects incorrect password → displays error message → allows retry (max 3 attempts) → locks account after 3 failed attempts. AF-2 (Account Locked): System detects locked account → displays message with unlock instructions → sends email with unlock link. AF-3 (Account Suspended): System detects suspended account → displays reason → provides contact information for support.
Special Requirements	Session timeout after 30 minutes of inactivity. Account lockout mechanism after failed login attempts. Support for institutional SSO integration.
Frequency of Use	High (Multiple times daily)

4.2.3 Student Dashboard Use Case

Table 12: View Student Dashboard

Use Case ID	UC-003
Use Case Name	View Student Dashboard
Priority	High
Trigger	Student logs into the system and accesses the personal performance dashboard.
Pre-Condition	Student is logged into the system. Student has an existing academic record in the system. Data synchronization with SIS/LMS is up to date.
Post-Condition	Student views personal risk status, performance indicators, and recommendations. Student privacy is maintained throughout.

Normal Flow	1. Student logs into the student portal. 2. System authenticates student and retrieves academic data. 3. System calculates current risk score based on available indicators. 4. System displays risk level, attendance rate, assignment completion, and quiz/exam performance. 5. System displays improvement recommendations based on risk factors. 6. Student reviews their academic standing.
Alternate Flow	AF-1 (No Data Available): No academic data has been synced yet → system displays a message indicating data is pending → prompts student to contact advisor. AF-2 (Session Timeout): Session expires during view → system prompts re-login → redirects back to dashboard.
Special Requirements	Students can only view their own data. Data must be refreshed within 24 hours of the latest academic records update.
Frequency of Use	Moderate (Several times per week)

4.2.4 Risk Classification Use Cases

Table 13: Risk Classification

Use Case ID	UC-004
Use Case Name	Risk Classification
Priority	Critical
Trigger	System receives updated academic data from the institutional database.
Pre-Condition	Academic data has been retrieved and validated. Risk threshold values have been configured by the administrator. Classification engine is operational.
Post-Condition	Each student is assigned a risk level (Low, Medium, or High). Risk classifications are stored and accessible to authorized stakeholders. Notifications are triggered if a risk escalation is detected.
Normal Flow	1. System retrieves latest academic indicators for each student. 2. System computes a weighted risk score: attendance (30%), assignment completion (20%), quiz performance (20%), examination performance (30%). 3. System compares score against configured thresholds.

	4. System assigns Low, Medium, or High risk level accordingly. 5. System updates the student record with the new risk classification. 6. If risk level has escalated, system triggers the alert notification use case.
Alternate Flow	AF-1 (Incomplete Data): One or more indicators are missing → system uses available data and flags incomplete record → alert advisor of data gap. AF-2 (Threshold Not Configured): Administrator has not set thresholds → system uses default values → logs warning for administrator.
Special Requirements	Classification must run automatically every 24 hours. Threshold values must be adjustable without system redeployment.
Frequency of Use	Critical (Automated, runs daily)

4.2.5 Advisor Alert Use Cases

Table 14: Send Alert Notification

Use Case ID	UC-005
Use Case Name	Send Alert Notification
Priority	Critical
Trigger	A student's risk level escalates (e.g., Low to Medium, or Medium to High).
Pre-Condition	Risk classification has been completed for the student. Advisor contact information is up to date in the system. Notification service is operational.
Post-Condition	Advisor receives an email and in-system notification about the student's escalated risk. Student receives a notification of their updated risk status. Notification is logged with timestamp in the audit trail.
Normal Flow	1. System detects risk level escalation after classification. 2. System identifies the assigned academic advisor for the student. 3. System composes notification message with student name, risk level, and contributing factors. 4. System sends email notification to advisor and student. 5. System creates in-system notification for the advisor's dashboard. 6. System logs notification delivery in the audit trail.

Alternate Flow	AF-1 (Notification Delivery Failure): Email fails to deliver → system retries after 5 minutes → if still fails, logs failure and flags for admin review. AF-2 (Advisor Not Assigned): No advisor is assigned to the student → system sends notification to department administrator → administrator assigns advisor.
Special Requirements	Notification must be sent within 24 hours of risk escalation detection. Both email and in-system notification channels must be supported.
Frequency of Use	Moderate to High (Triggered by risk escalation events)

4.2.6 Intervention Logging Use Case

Table 15: Log Intervention

Use Case ID	UC-006
Use Case Name	Log Intervention
Priority	High
Trigger	Advisor completes an intervention meeting or action with an at-risk student.
Pre-Condition	Advisor is logged into the system. Student has an active risk record in the system. Advisor is authorized to log interventions for the student.
Post-Condition	Intervention record is saved in the student's academic history. Intervention details are accessible to authorized stakeholders. Intervention count is updated in the reporting module.
Normal Flow	1. Advisor navigates to the student's profile. 2. Advisor selects 'Log Intervention'. 3. System displays the intervention entry form. 4. Advisor enters intervention date, type (e.g., meeting, email, referral), notes, and outcome. 5. System validates and saves the intervention record. 6. System confirms successful logging to the advisor.
Alternate Flow	AF-1 (Incomplete Form): Advisor submits form with missing required fields → system highlights fields → advisor completes and resubmits. AF-2 (Session Expired): Session expires mid-entry → system prompts re-login → form data is temporarily preserved.
Special Requirements	Intervention records are accessible only to the assigned advisor and administrators. All records are time-stamped and immutable after submission.

Frequency of Use	Moderate (Multiple times per week during academic term)
-------------------------	---

4.2.7 Report Generation Use Case

Table 16: Generate Report

Use Case ID	UC-007
Use Case Name	Generate Report
Priority	High
Trigger	Administrator requests a performance report or scheduled monthly report generation is triggered.
Pre-Condition	Administrator is logged into the system. Academic data is available and up to date in the system. Reporting module is operational.
Post-Condition	Report is generated and delivered to the administrator. Report is saved in the system for future reference. Report generation event is logged in the audit trail.
Normal Flow	1. Administrator navigates to the Reports section. 2. Administrator selects report type (monthly summary, risk distribution, intervention effectiveness). 3. Administrator specifies date range and filters. 4. System retrieves and aggregates relevant data. 5. System generates the report in PDF or Excel format. 6. System delivers report to administrator and saves a copy in the system.
Alternate Flow	AF-1 (No Data for Period): No academic data exists for the selected range → system displays informational message and suggests an adjusted range. AF-2 (Report Generation Timeout): Report takes too long → system notifies administrator and emails report upon completion.
Special Requirements	Monthly reports must be generated automatically on the first of each month. Reports must support PDF and Excel export formats.
Frequency of Use	Moderate (On-demand plus automated monthly generation)

4.2.8 System Configuration Use Case

Table 17: Configure System Thresholds

Use Case ID	UC-008
Use Case Name	Configure System Thresholds
Priority	High
Trigger	Administrator accesses the system configuration panel to adjust risk parameters.
Pre-Condition	Administrator is logged into the system with configuration privileges. System configuration module is accessible. Current threshold values are displayed.
Post-Condition	New threshold values are saved and applied to subsequent risk classifications. Configuration change is logged in the audit trail. Administrator receives confirmation of successful update.
Normal Flow	1. Administrator navigates to the System Configuration panel. 2. Administrator views current threshold and weighting settings. 3. Administrator modifies threshold values (e.g., Low/Medium/High risk boundaries) or indicator weights. 4. System validates that all weights sum to 100% and threshold ranges do not overlap. 5. System saves the new configuration. 6. System applies the new configuration to the next classification cycle. 7. System confirms successful update to administrator.
Alternate Flow	AF-1 (Invalid Configuration): Weights do not sum to 100% or ranges overlap → system displays validation error → administrator corrects values. AF-2 (Unauthorized Access): Non-administrator attempts to access configuration → system denies access and logs the attempt.
Special Requirements	Configuration changes take effect from the next scheduled classification run. All changes are logged with administrator ID and timestamp.
Frequency of Use	Low (At semester start or when policy changes occur)

4.3 System Sequence Diagrams (SSD)

The following System Sequence Diagrams illustrate the interactions between external actors and the system boundary for key use cases. All SSDs were created using draw.io following UML notation discussed in lectures.

4.3.1 Student Registration SSD

This diagram shows the interaction between the user and the system during the registration process, including data validation, account creation, and confirmation notification. The registration process is illustrated as shown in Figure 2.

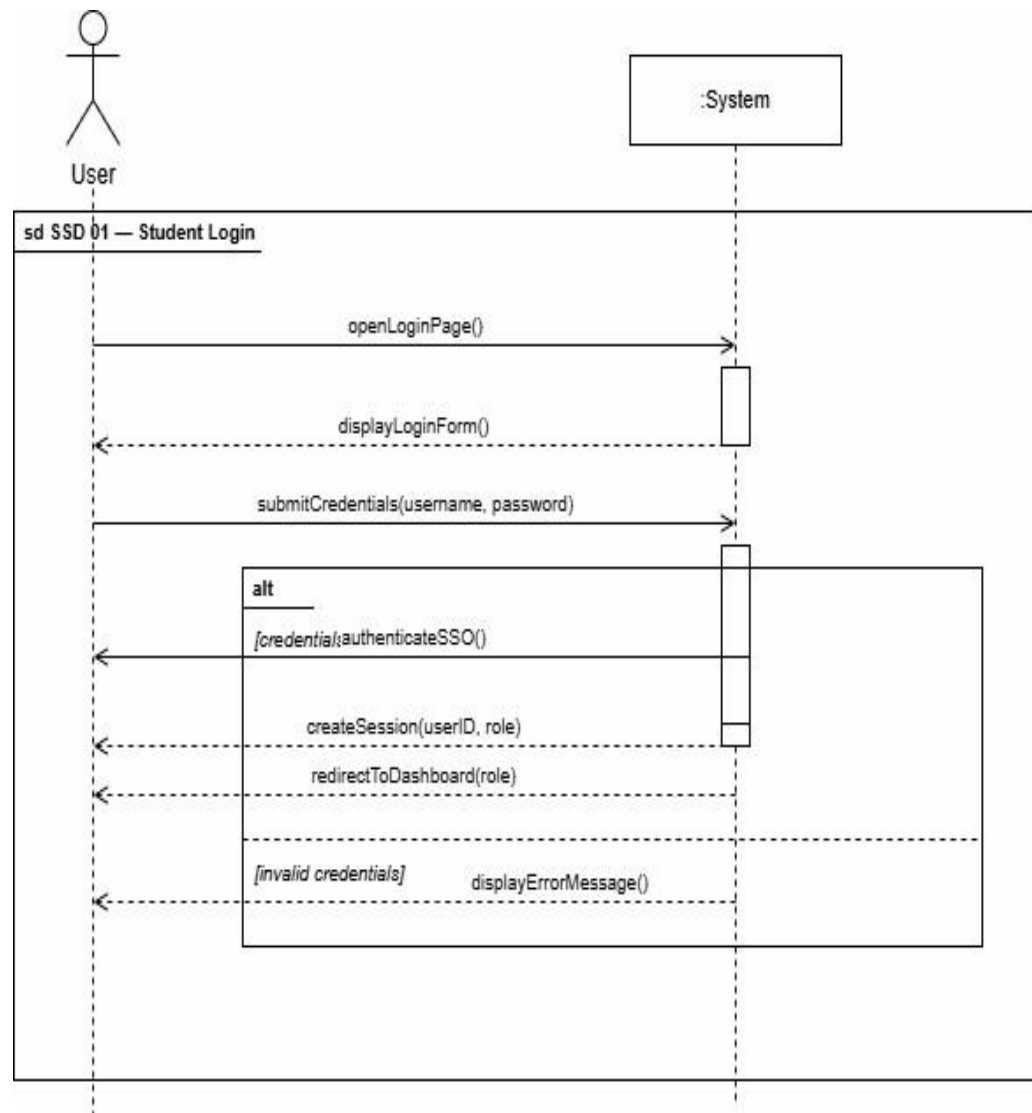


Figure 2: Student Registration SSD

4.3.2 User Login SSD

This diagram illustrates the authentication flow between the user and the system, covering credential validation, session creation, and role-based dashboard redirection, as shown in Figure 3.

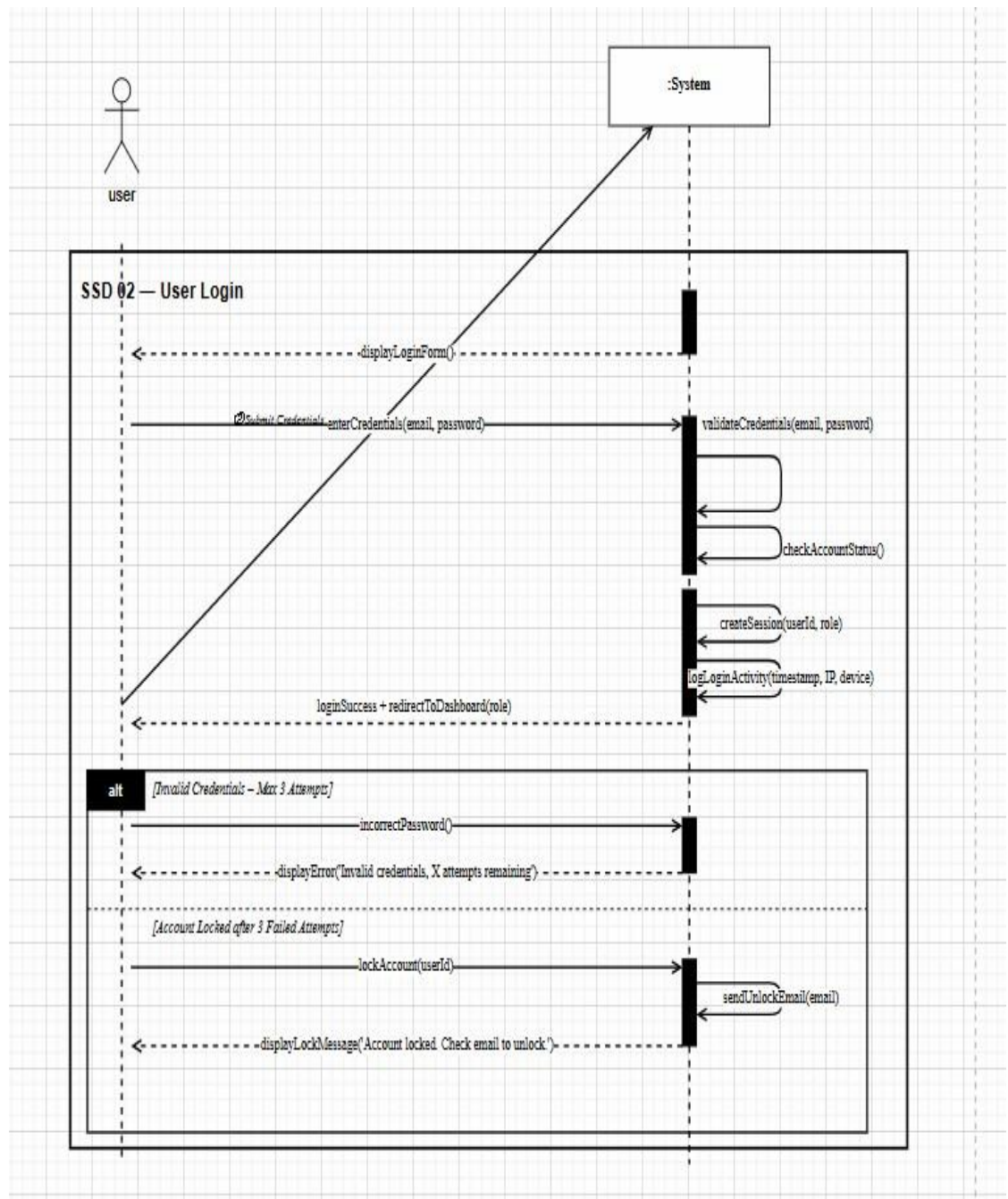


Figure 3: User Login SSD

4.3.3 Risk Assessment SSD

This diagram shows the automated interaction between the system components during the risk classification cycle. It covers data retrieval, score computation, threshold comparison, and risk level assignment, as shown in Figure 4.

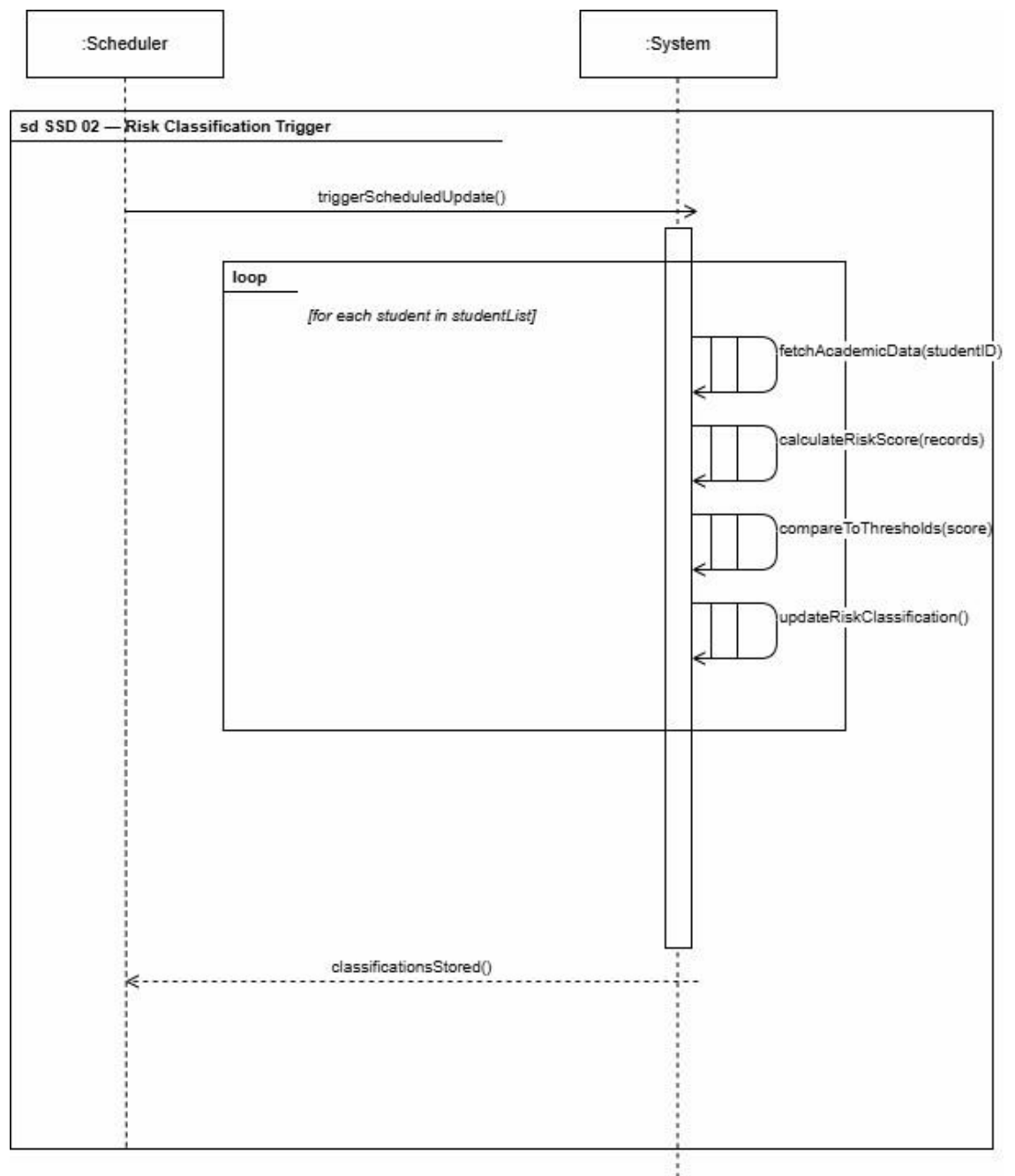


Figure 4: Risk Assessment SSD

4.3.4 Alert Notification SSD

This diagram illustrates the system interaction triggered when a student's risk level escalates. It covers risk detection, advisor lookup, notification composition, and delivery confirmation, as shown in Figure 5.

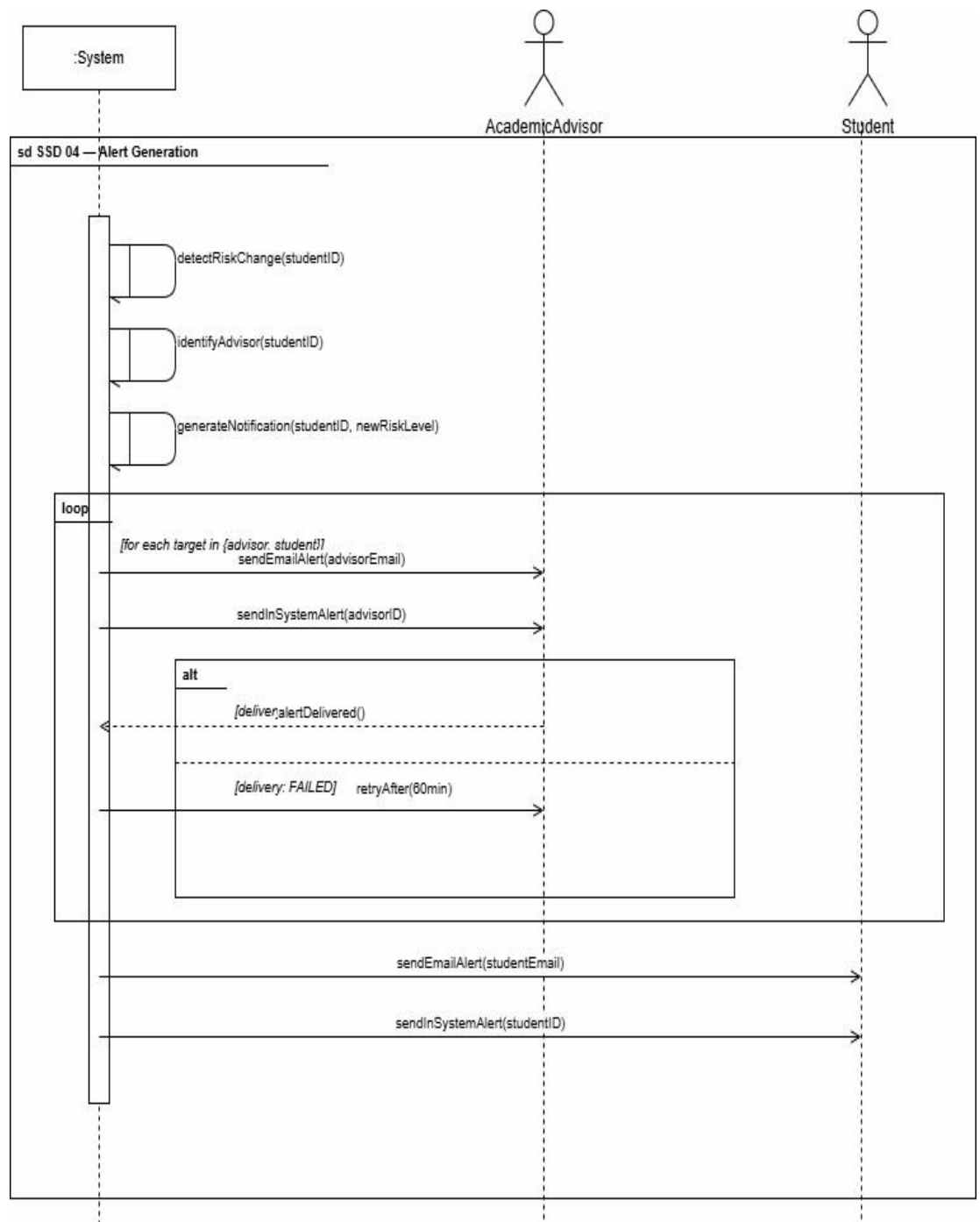


Figure 5: Alert Notification SSD

4.3.5 Advisor Dashboard SSD

This diagram shows how the academic advisor accesses the dashboard. The system authenticates the advisor, retrieves the list of assigned at-risk students, and displays the updated risk dashboard along with pending alerts, as shown in Figure 6.

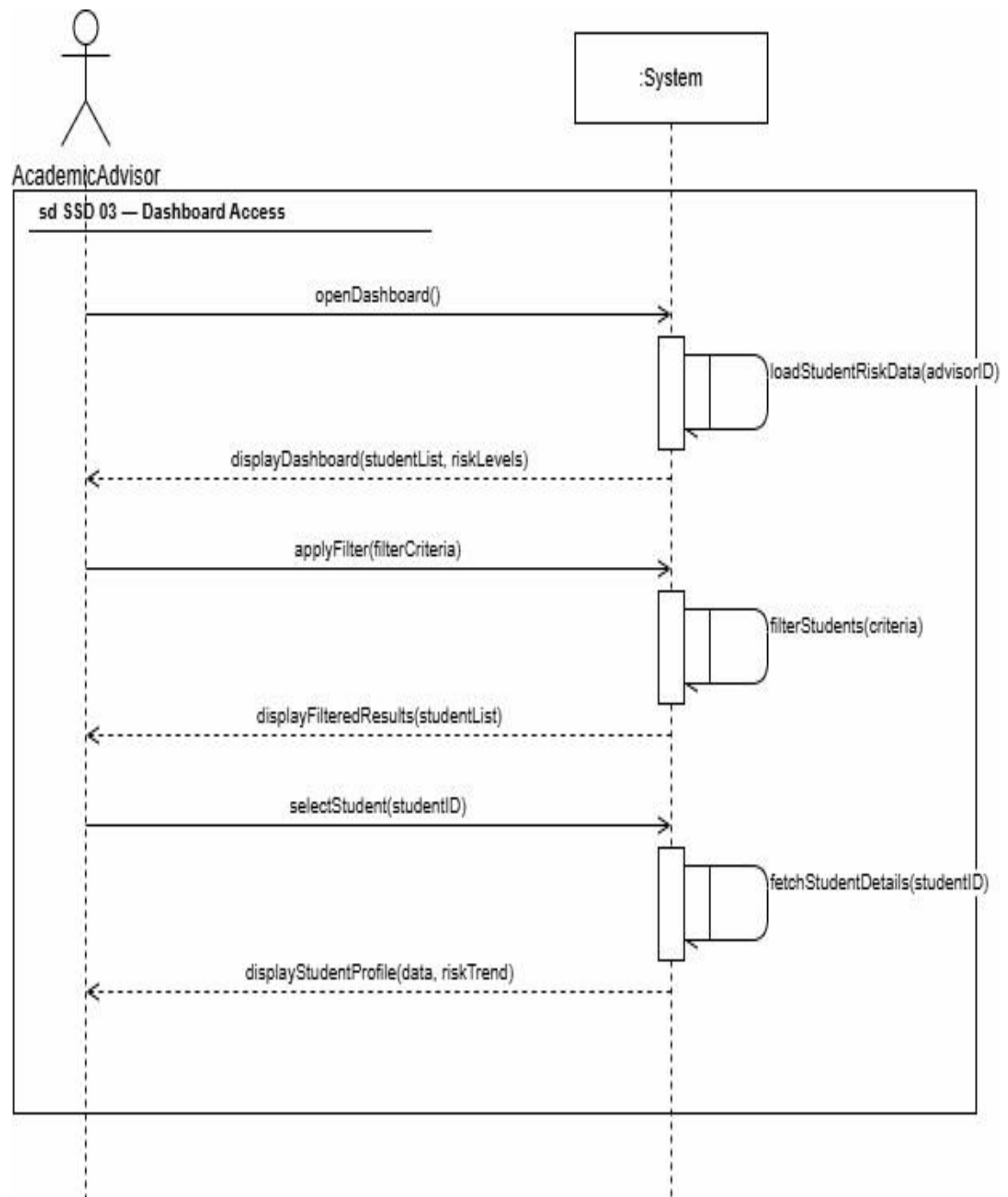


Figure 6: Advisor Dashboard SSD

4.3.6 Intervention Logging SSD

This diagram demonstrates the interaction between the academic advisor and the system during the intervention logging process, ensuring that all records are properly validated and stored in the student's academic history, as shown in Figure 7.

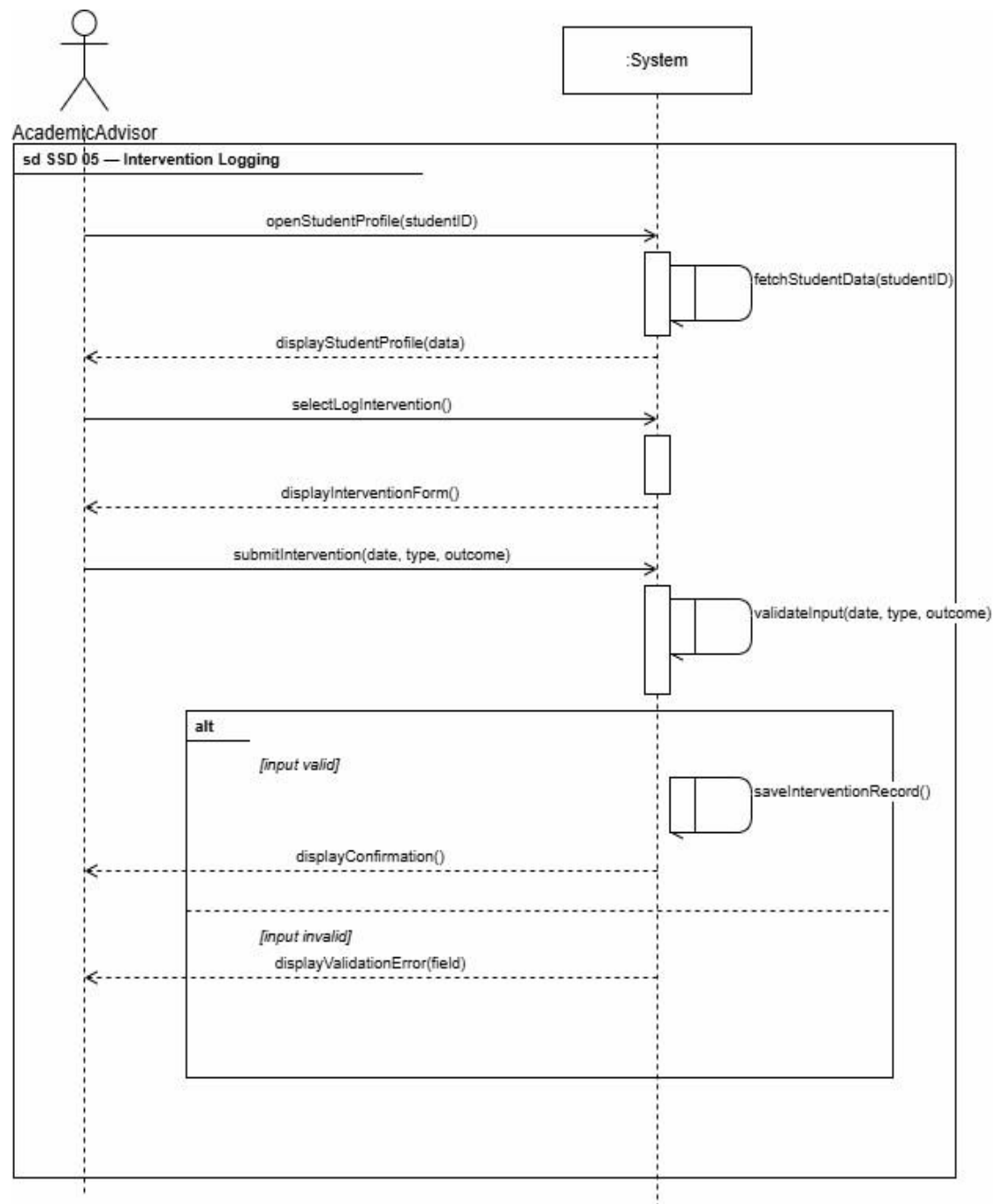


Figure 7: Intervention Logging SSD

4.3.7 Report Generation SSD

This diagram illustrates the interaction between the administrator and the system for generating academic performance reports, covering data aggregation, report compilation, and delivery, as shown in Figure 8.

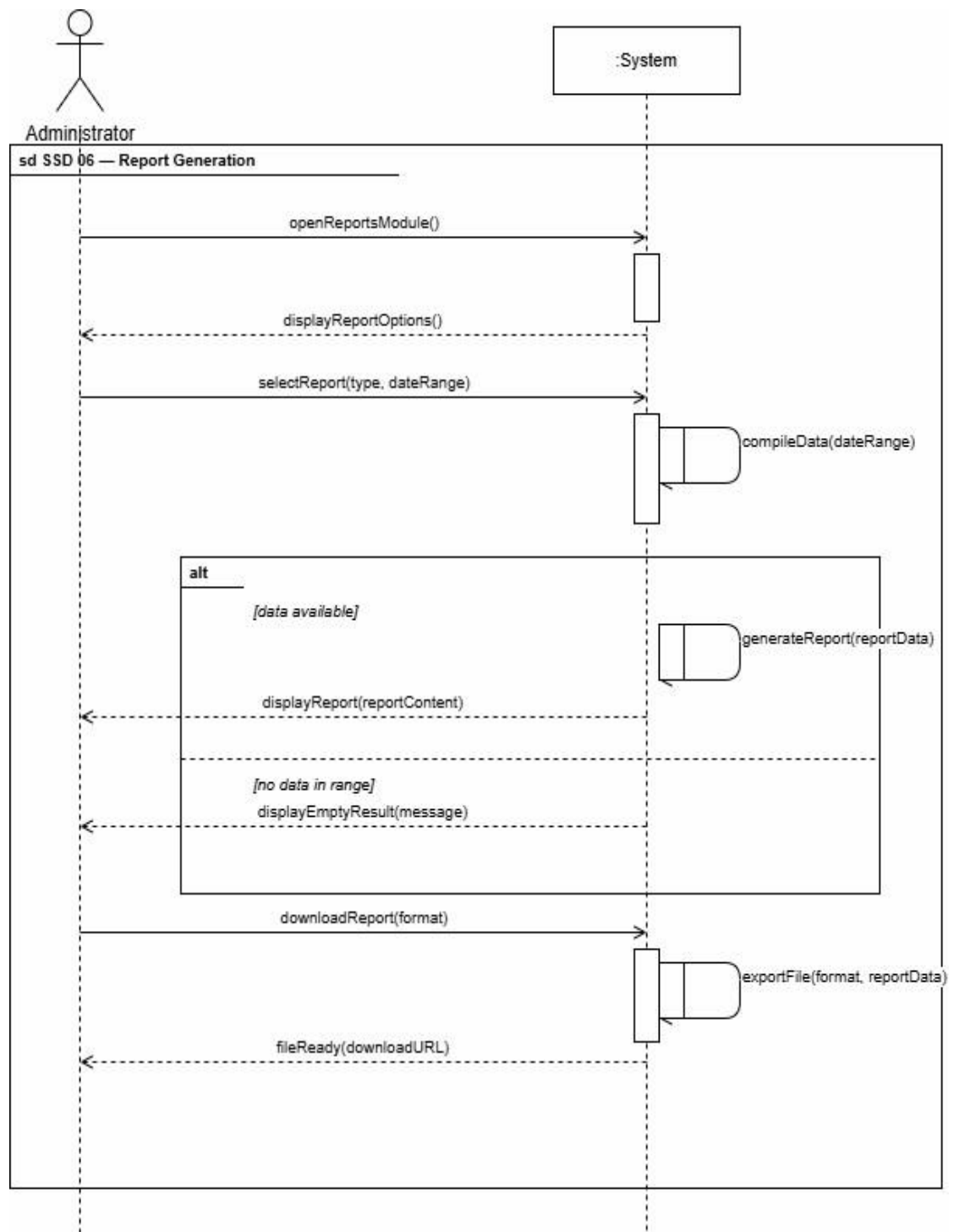


Figure 9: System Configuration SSD

5 Design Phase Artifacts

5.1 Domain Model

This domain model represents the key entities and relationships in the Academic Performance Early-Warning System. It defines classes such as User, Student, Advisor, Faculty, Administrator, and their associations with other entities like CourseEnrollment, PerformanceRecord, RiskAssessment, AlertNotification, and InterventionLog.

It visually maps one-to-one, one-to-many, and many-to-one relationships, ensuring data consistency and traceability across academic performance monitoring workflows. The domain model is shown in Figure 10.

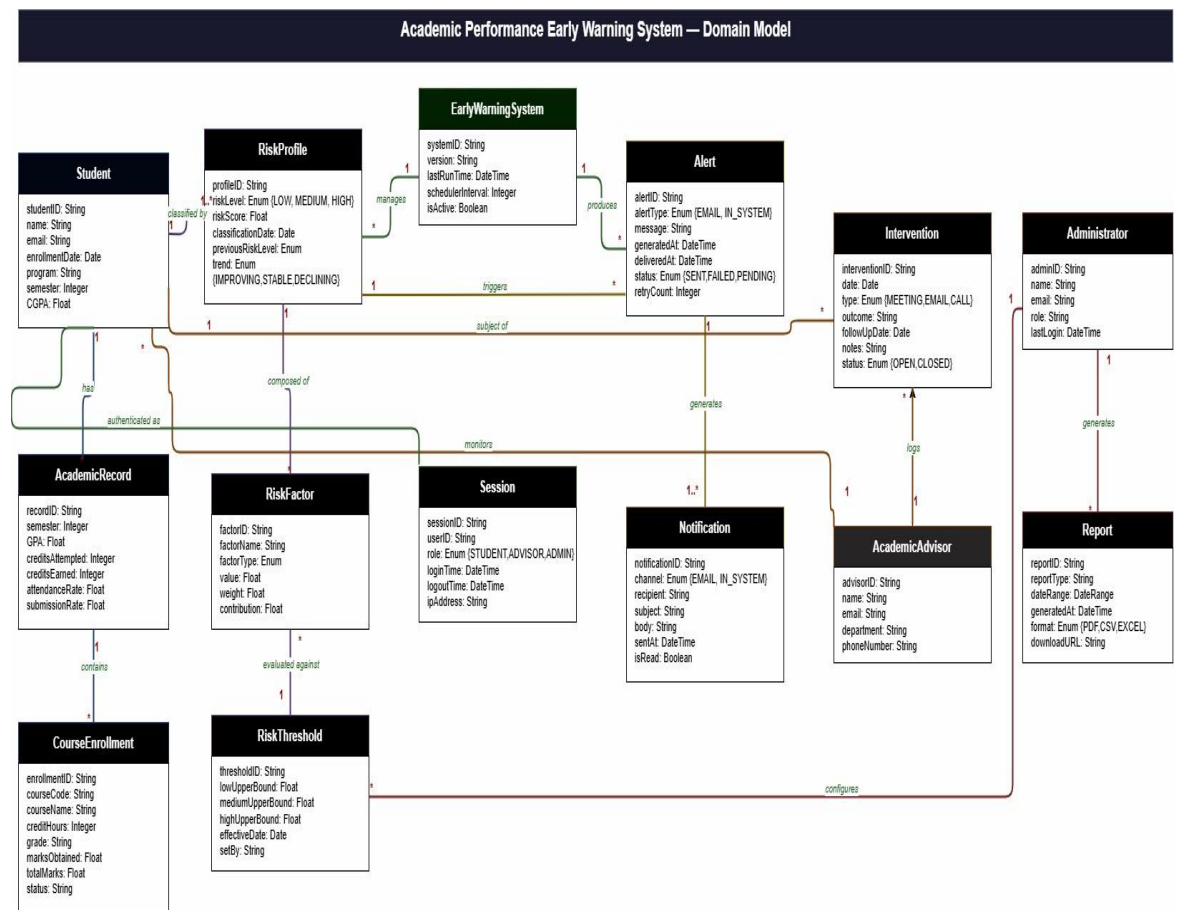


Figure 10: Domain Model

5.2 Sequence Diagram

The Complete System Sequence Diagram illustrates the end-to-end student academic monitoring journey through the Academic Performance Early-Warning System, from initial registration to intervention logging and report generation. The diagram demonstrates interactions between four primary actors (Student, Academic Advisor, Faculty Member, and System Administrator) and key system components including the authentication service, data ingestion engine, risk classification engine, notification service, and reporting module.

The sequence begins with user authentication where students and advisors register and log into the system. Academic data is periodically ingested from the institutional SIS and LMS. The Risk Classification Engine processes attendance, assignment, quiz, and examination data to compute a risk score and assign a risk category. When risk escalation is detected, the notification service dispatches email and in-system alerts to the assigned advisor and the student.

During the intervention phase, the advisor reviews the student's risk dashboard, reviews contributing performance factors, and logs the intervention outcome. Department administrators access aggregate reports through the reporting module, which consolidates risk distribution, intervention counts, and performance trends across all enrolled students. The system maintains a comprehensive audit trail throughout the process and generates automated monthly reports, as shown in Figure 11.

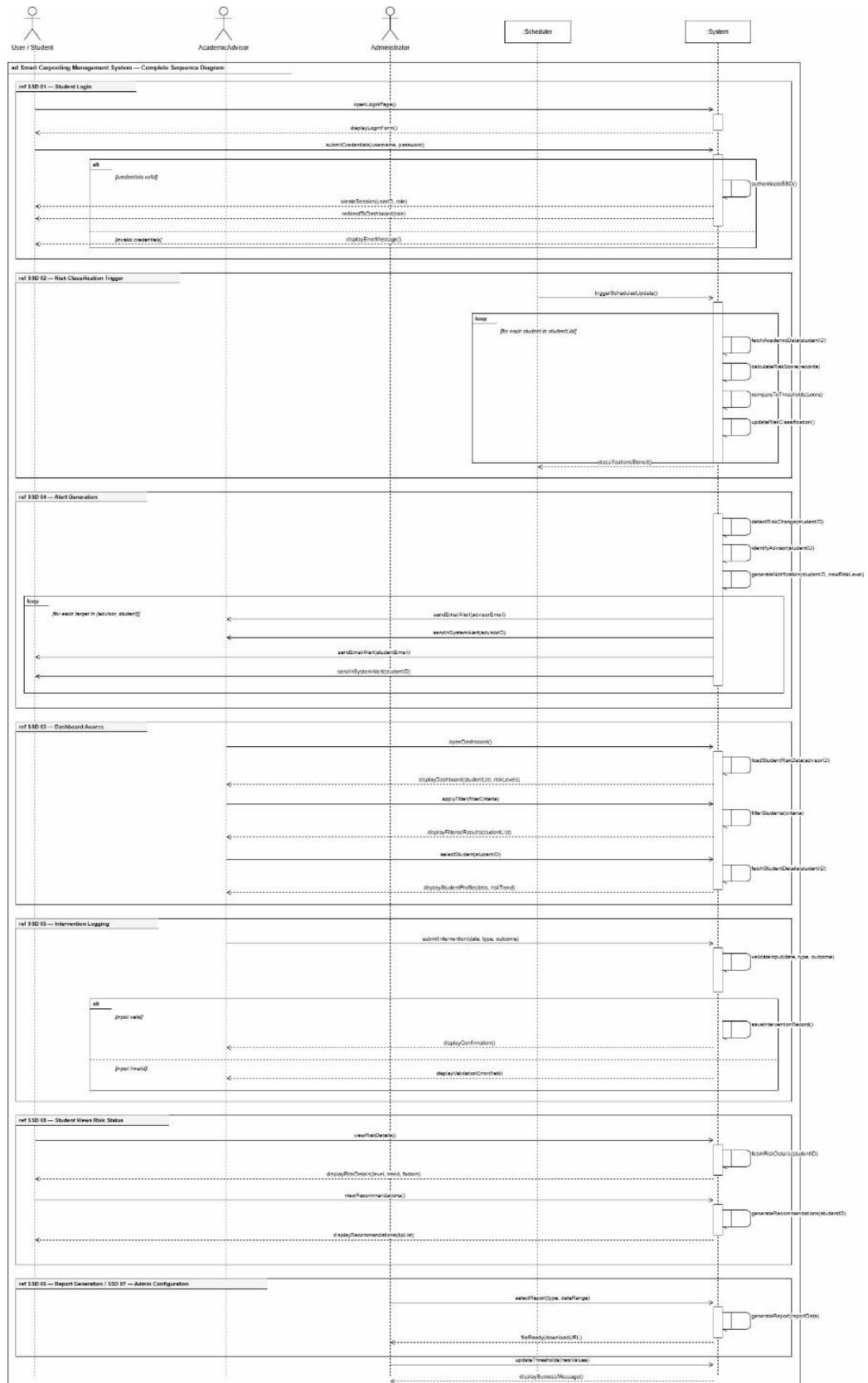


Figure 11: Complete System Sequence Diagram

6 Detailed Design & Modeling Artifacts

This chapter presents the detailed design and modeling artifacts for the Academic Performance Early-Warning System. It includes System Sequence Diagrams (SSDs) that were developed in the Requirements Analysis phase, along with comprehensive Operation Contracts that specify the preconditions, postconditions, and system state changes for each major operation.

6.1 System Sequence Diagrams (SSD)

The System Sequence Diagrams (SSDs) illustrate the interactions between actors and the system for specific use cases. These diagrams show the sequence of messages exchanged, including user inputs, system processing, and system responses. Please refer to Section 4.3 for the complete set of SSDs.

6.2 Operation Contracts

Operation contracts formally specify the system behavior for each operation. They define the preconditions that must be true before an operation can execute and the postconditions that describe the state changes after successful execution. The following contracts correspond to the major system operations identified in the use cases and SSDs.

Contract CO-01

Operation: registerUser(name: String, email: String, userId: String, password: String, role: String)

Cross References: Use Cases: UC-001 (User Registration)

Preconditions:

- Email is not already registered
- All required fields are provided
- Email format is valid
- Password meets security requirements
- Role is one of: Student, Advisor, Faculty, Administrator

Postconditions:

- New User instance created

- userId auto-generated
- User data stored securely (password hashed)
- Role assigned
- Account status set to Active
- Verification email sent

Contract CO-02

Operation: loginUser(email: String, password: String)

Cross Reference: UC-002 (User Login)

Preconditions:

- User exists
- Account is active
- Credentials provided

Postconditions:

- If valid:
 - Session created
 - User authenticated
 - Login timestamp updated
 - Redirect to dashboard
- If invalid:
 - Error message shown
 - No session created

Contract CO-03

Operation: viewDashboard(userId: int)

Cross Reference: UC-003 (Student Dashboard)

Preconditions:

- User is authenticated
- Academic data available

Postconditions:

- Risk score calculated
- Risk level displayed (Low/Medium/High)
- Performance indicators shown: Attendance, Assignments, Quiz & Exam scores
- Recommendations displayed

Contract CO-04

Operation: calculateRisk(studentId: int)

Cross Reference: UC-004 (Risk Classification)

Preconditions:

- Academic data available
- Thresholds configured

Postconditions:

- Risk score computed:
 - Attendance (30%)
 - Assignments (20%)
 - Quiz (20%)
 - Exam (30%)
- Risk level assigned
- Student record updated
- If risk increased → alert triggered

Contract CO-05

Operation: sendAlert(studentId: int, riskLevel: String)

Cross Reference: UC-005 (Alert Notification)

Preconditions:

- Risk escalation detected
- Advisor assigned

Postconditions:

- Notification created
- Email sent to Advisor and Student
- Notification logged
- Alert displayed in dashboard

Contract CO-06

Operation: viewAdvisorDashboard(advisorId: int)

Cross Reference: UC-005 / Advisor Dashboard

Preconditions:

- Advisor authenticated

Postconditions:

- List of assigned students retrieved
- Risk levels displayed
- Alerts displayed
- Filtering options enabled

Contract CO-07

Operation: logIntervention(studentId: int, advisorId: int, details: String)

Cross Reference: UC-006 (Intervention Logging)

Preconditions:

- Advisor logged in
- Student exists
- Student has risk record

Postconditions:

- New intervention record created
- Timestamp added
- Record stored permanently
- Confirmation sent

Contract CO-08

Operation: generateReport(adminId: int, type: String, dateRange: Date)

Cross Reference: UC-007 (Report Generation)

Preconditions:

- Admin authenticated
- Data available

Postconditions:

- Data aggregated
- Report generated (PDF/Excel)
- Report stored
- Report delivered

Contract CO-09

Operation: configureThresholds(adminId: int, thresholds: Object)

Cross Reference: UC-008 (System Configuration)

Preconditions:

- Admin authenticated
- Valid threshold values

Postconditions:

- Thresholds updated
- Changes saved
- Applied to next risk cycle
- Audit log updated

7 Structural Design

This chapter presents the structural design of the Academic Performance Early-Warning System through Class Diagrams and Package Diagrams. These diagrams provide a static view of the system architecture, showing the organization of classes, their attributes, methods, relationships, and the overall package structure.

7.1 Class Diagram

The Class Diagram illustrates the static structure of the system, showing classes, attributes, operations, and relationships.

Core Classes include:

- User (base class)
- Student, Advisor, Faculty, Administrator (derived classes)
- PerformanceRecord
- RiskAssessment
- AlertNotification
- InterventionLog
- Report

Key Features:

- Inheritance between User and specialized roles
- Associations between Student and PerformanceRecord
- Relationships between Advisor and InterventionLog
- Encapsulation of data and operations

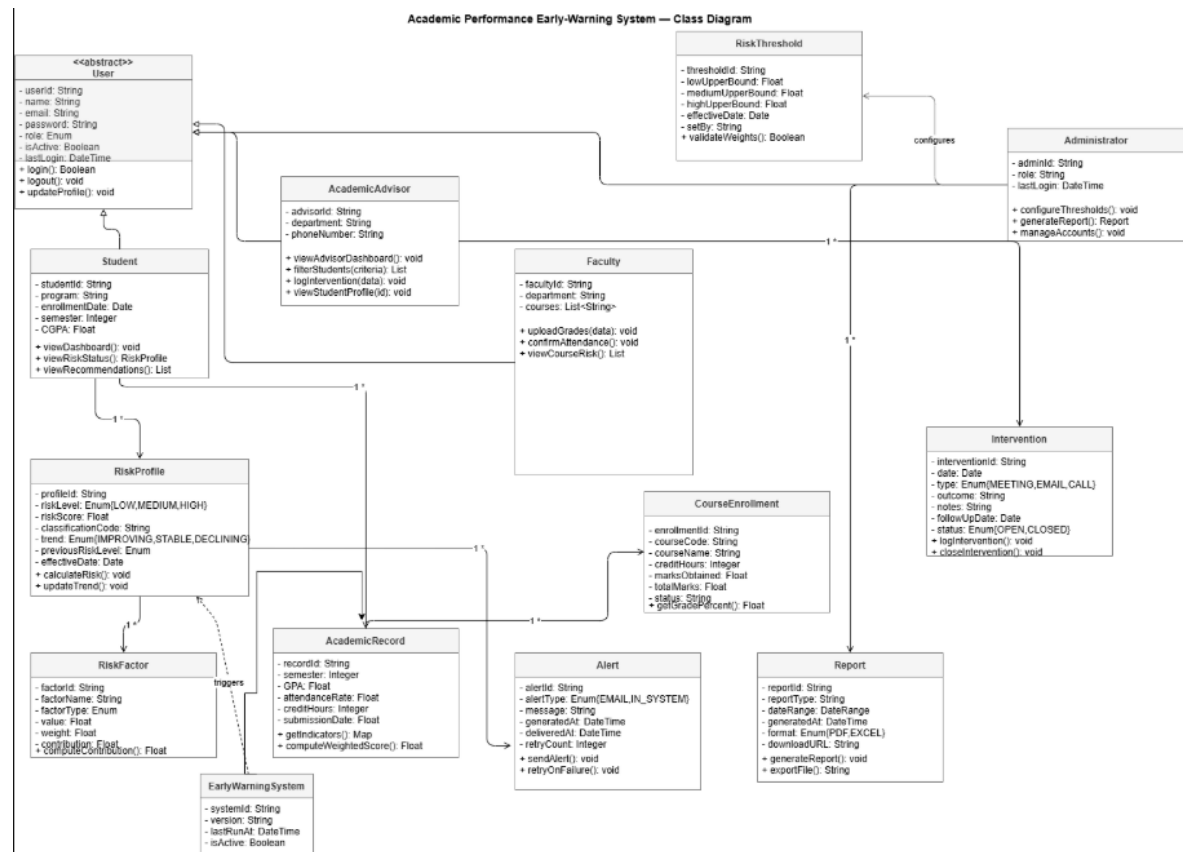


Figure 7.1: Class Diagram showing system entities and relationships

7.2 Package Diagram

The Package Diagram organizes the system into logical packages, showing the dependencies between different layers and components. The system follows a layered architecture pattern with clear separation of concerns.

Package Structure:

- Presentation Layer: DashboardView, LoginView, ReportView
- Application Layer: AuthenticationService, RiskService, NotificationService, ReportingService
- Domain Layer: User, Student, RiskAssessment, InterventionLog
- Data Layer: UserRepository, StudentRepository, RiskRepository
- External Services: SIS/LMS Integration, Email Service

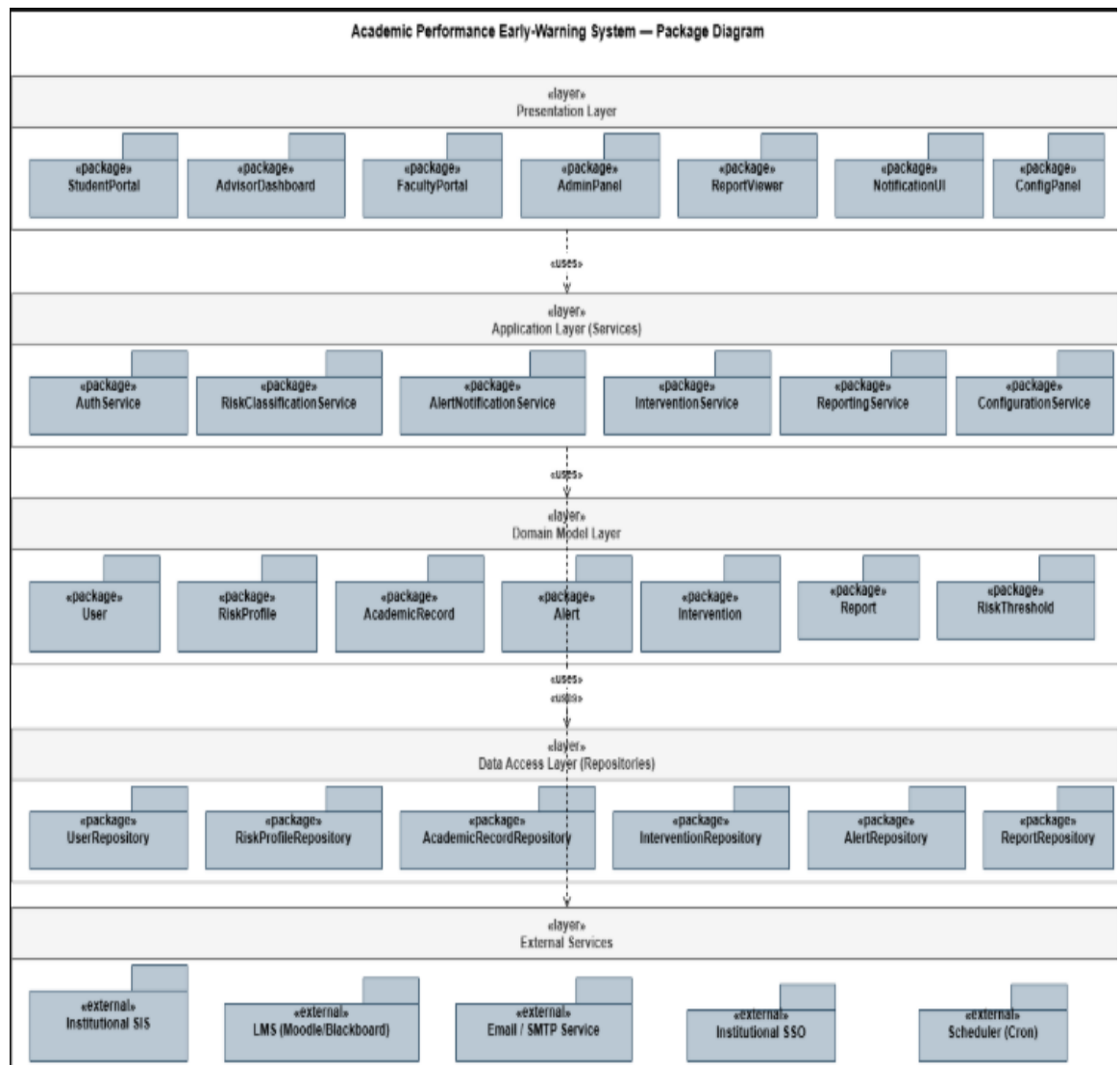


Figure 7.2: Package Diagram showing layered architecture

8 Interaction Design

This chapter illustrates the interaction design through Communication Diagrams, which show how objects collaborate to realize a particular use case or scenario. The diagram emphasizes the structural organization of objects that interact through message passing.

8.1 Communication Diagram

The Communication Diagram presents the interaction between objects in the 'Risk Monitoring & Intervention' scenario. It shows the links between objects and the messages that pass along these links, numbered in sequence order to indicate the flow of control.

Scenario: Risk Monitoring & Intervention

Participating Objects:

- :Student
- :Advisor
- :DashboardView
- :RiskService
- :NotificationService
- :InterventionService

Flow:

- Student logs in
- System calculates risk
- RiskService updates level
- NotificationService sends alert
- Advisor reviews dashboard
- Advisor logs intervention

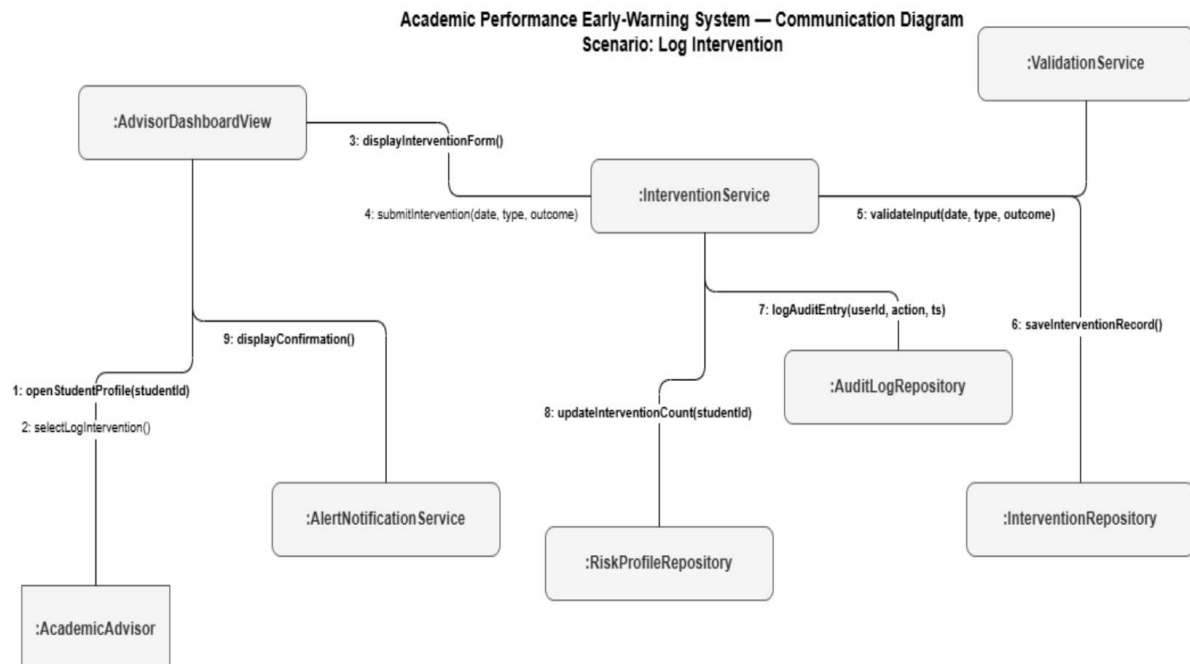


Figure 8.1: Communication Diagram for Risk Monitoring & Intervention scenario

9 Behavioral Modeling

This chapter presents behavioral modeling diagrams that capture the dynamic aspects of the system. State Machine Diagrams show the states of objects and the transitions between states, while Activity Diagrams model the workflow of business processes.

9.1 State Machine Diagram

The State Machine Diagram models the lifecycle of a Risk Assessment, showing all possible states and the events that cause transitions between states. This diagram is essential for understanding the risk monitoring workflow and ensuring proper state management.

States:

- Initial
- Low Risk
- Medium Risk
- High Risk
- Intervention Ongoing
- Improved
- Completed

Transitions:

- Data Update → Risk Recalculated
- Risk Increase → Alert Triggered
- Intervention → Risk Reduced

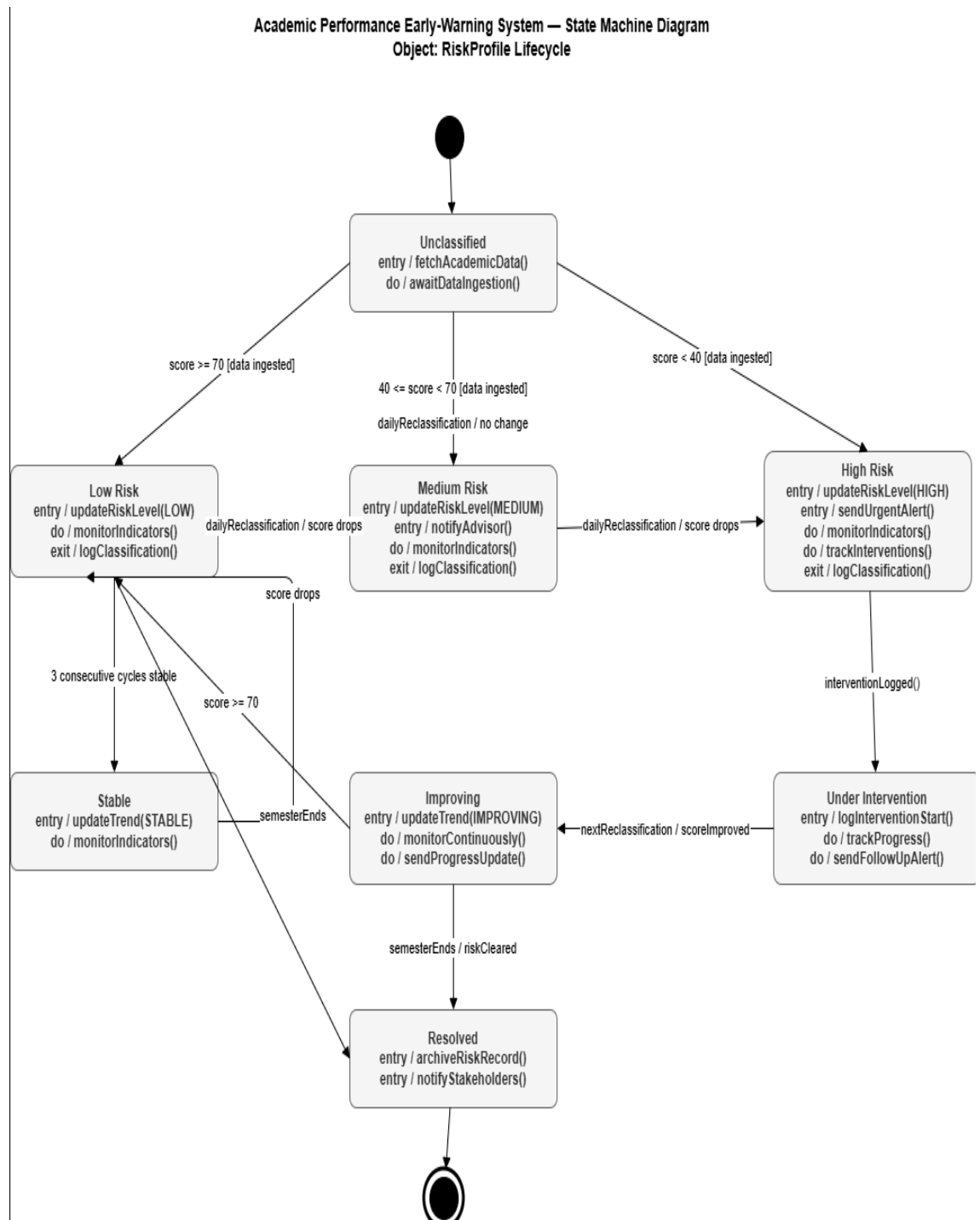


Figure 9.1: State Machine Diagram for Risk Lifecycle

9.2 Activity Diagram

The Activity Diagram models the workflow of the ‘Risk Monitoring Process’, showing the sequence of activities performed by different actors (Student, System, Advisor) and the control flow between them. Swimlanes are used to clearly delineate responsibilities.

Process Flow:

- Collect academic data
- Calculate risk score
- Decision: Risk level?
- If High → Send Alert
- Advisor reviews case
- Intervention logged
- Monitor improvement

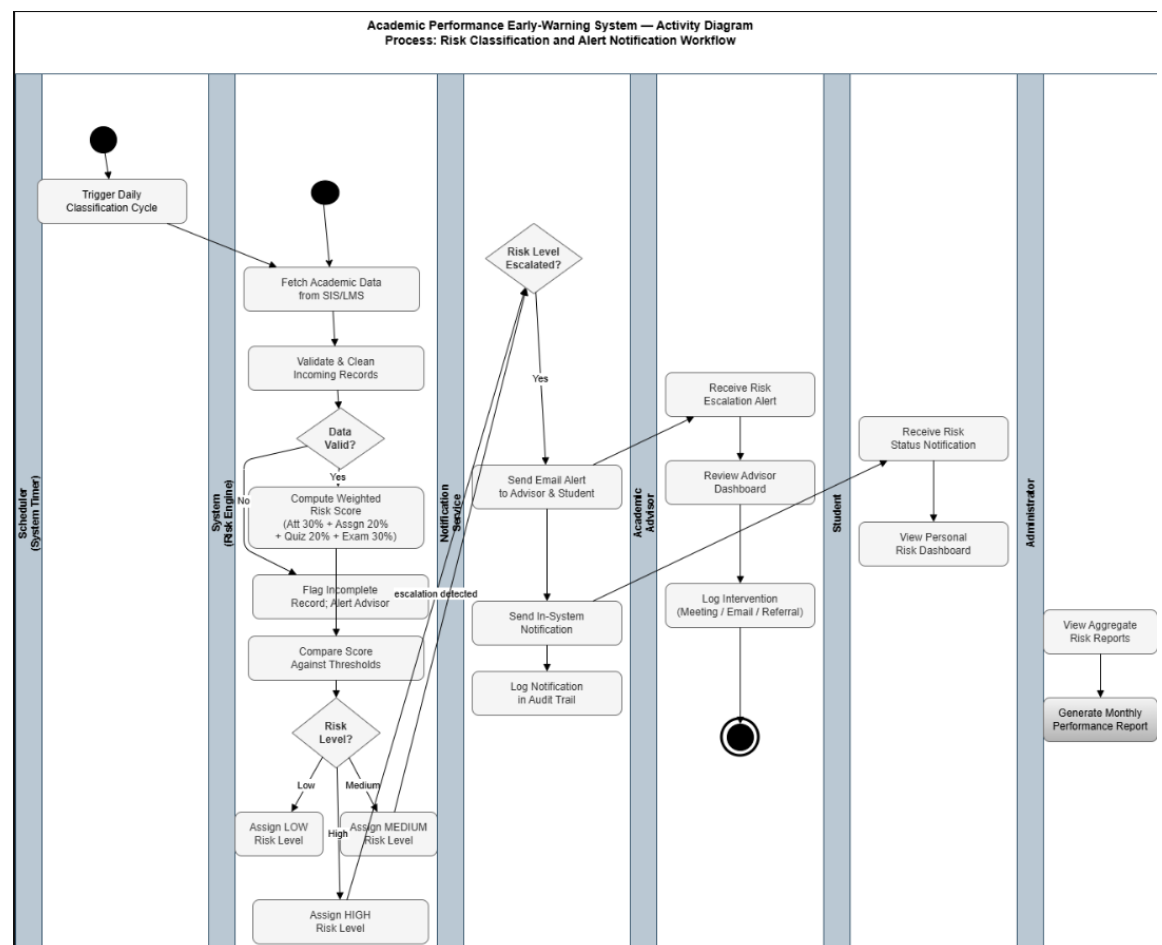


Figure 9.2: Activity Diagram for Risk Monitoring Process

10 Data Flow Modeling

This chapter presents the Data Flow Diagram (DFD) which models how data moves through the system. The DFD shows the processes, data stores, external entities, and the flow of information between them.

10.1 Data Flow Diagram (DFD)

The Level 0 DFD (Context Diagram) provides a high-level overview of the Academic Performance Early-Warning System, showing the system as a single process and its interactions with external entities. This diagram helps understand the system boundary and the inputs/outputs of the system.

External Entities:

- Student: End user who views their own performance indicators and receives notifications
- Academic Advisor: Monitors student risk dashboards and logs interventions
- Faculty Member: Provides course-level academic data inputs
- Administrator: Configures system thresholds and views aggregate reports
- SIS/LMS: External academic data sources providing attendance and grade information
- Email Service: External notification delivery system

Major Data Flows:

- Academic Data: SIS/LMS → System
- Risk Results: System → Dashboard
- Alerts: System → Advisor & Student
- Intervention Data: Advisor → System
- Reports: System → Administrator
- Configuration Updates: Administrator → System

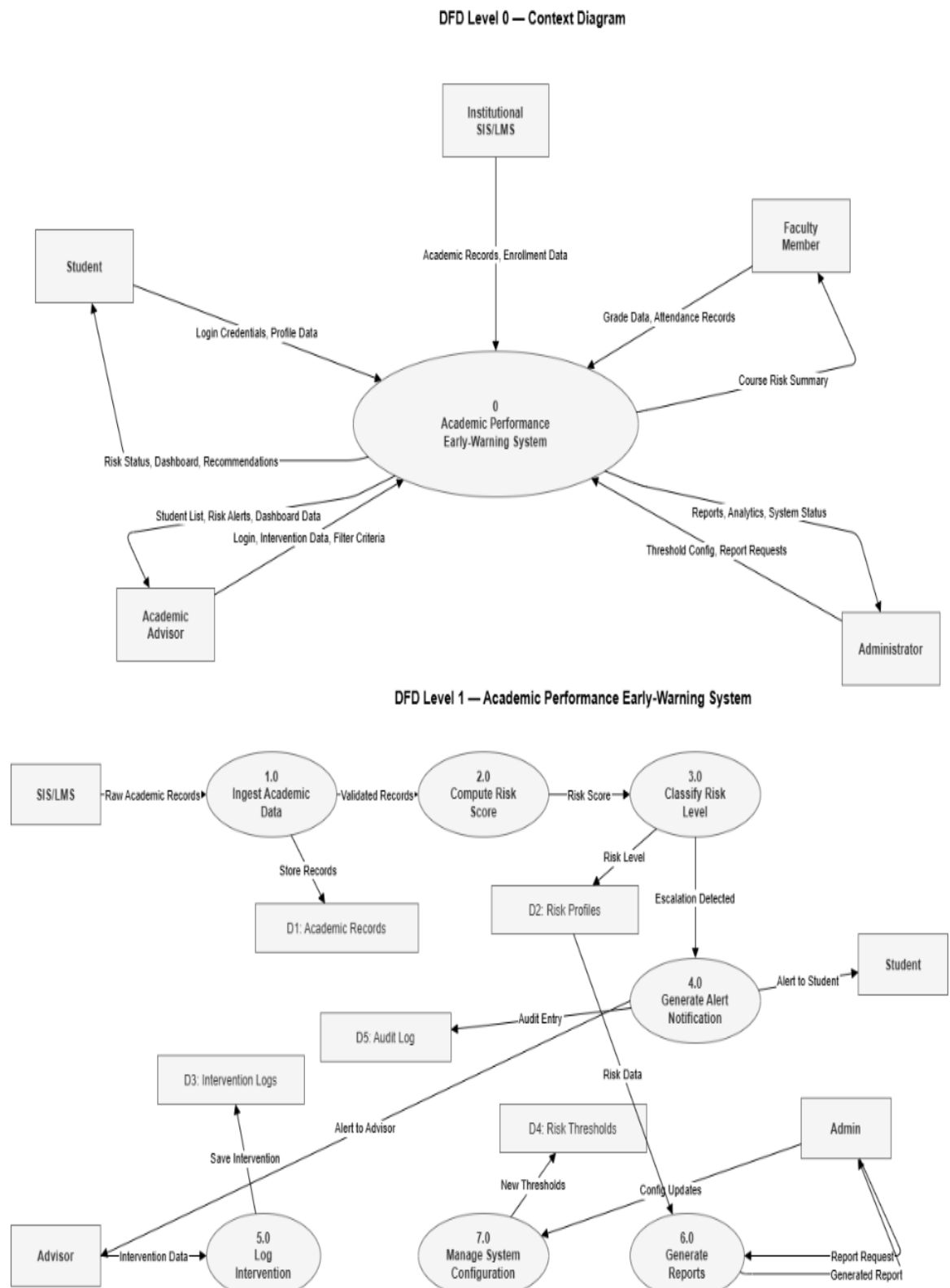


Figure 10.1: Data Flow Diagram Level 0 (Context Diagram)